

DA6701

Assignment V: Goal-Based Portfolio Optimisation

Team 14
May 2, 2026

1 Introduction

The exercise is a goal-based investing problem. A client saves ₹20,000 per month (with a 4 % annual step-up) over a 20-year horizon and wants to retire with a terminal corpus of ₹1.5 Cr. Three intermediate goals must be met along the way; any shortfall is borrowed at an exact 12 % effective annual rate. We must pick a single static allocation across five Indian equities — TCS, HDFCBANK, RELIANCE, SUNPHARMA, ITC — that maximises the probability of clearing the terminal goal.

Two completely different goal sequences are evaluated:

- **Sequence A (early-loaded):** ₹15 L at Y3, ₹25 L at Y7, ₹30 L at Y12.
- **Sequence B (back-loaded):** ₹10 L at Y8, ₹20 L at Y12, ₹40 L at Y16.

The optimal allocation is found in two ways: a brute-force search over a discrete weight grid, and a continuous bonus optimisation that allows short-selling.

2 Methodology

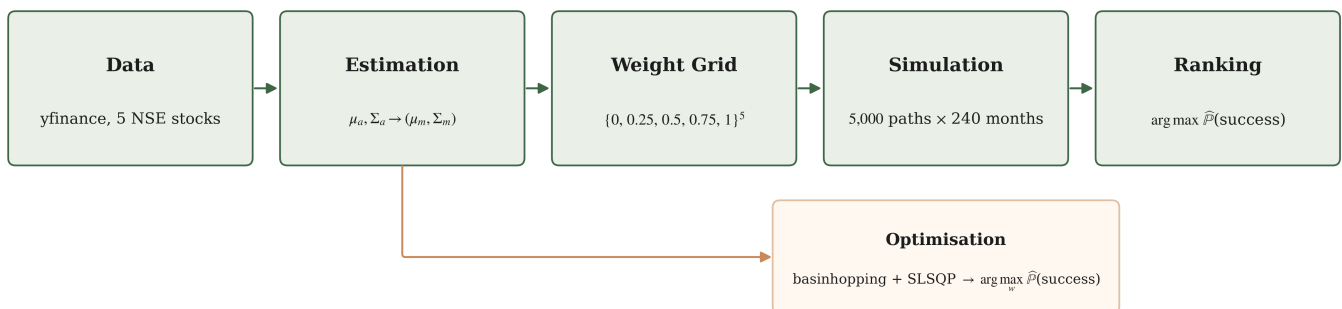


Figure 1: End-to-end methodology pipeline. The five mandatory steps (sage) run left to right; the optional bonus optimisation (peach) branches off after Estimation.

Figure 1 summarises the full pipeline. Daily prices are pulled with `yfinance`, daily log-returns are converted to annualised μ, Σ (and rescaled to monthly stats by dividing by 12), every static weight on the discrete grid is fed into a 5,000-path Monte-Carlo over 240 months, and the portfolio with the highest probability of reaching ₹1.5 Cr is reported. The bonus repeats the search with continuous weights and short-selling.

Modelling method:

- Time grid is monthly (240 steps).
- In each month: contribute, apply growth, accrue debt interest, deduct any due goal (borrow only the gap), then sweep any positive cash to the loan before reinvesting.
- Portfolio gross return per month is lognormal, $r_p = \exp(z)$, $z \sim \mathcal{N}(w^\top \mu_m, w^\top \Sigma_m w)$.
- Borrow rate 0.12 is converted to a monthly equivalent $(1.12)^{1/12} - 1 \approx 0.949\%$.
- Success means $A_{240} - D_{240} \geq 1.5 \text{ Cr.}$

3 Data and statistics

Adjusted daily closes from 1-Jan-2014 to 31-Dec-2023 give 2,465 trading days per ticker. Figure 2 shows the cumulative growth of ₹100 invested in each name on day one.

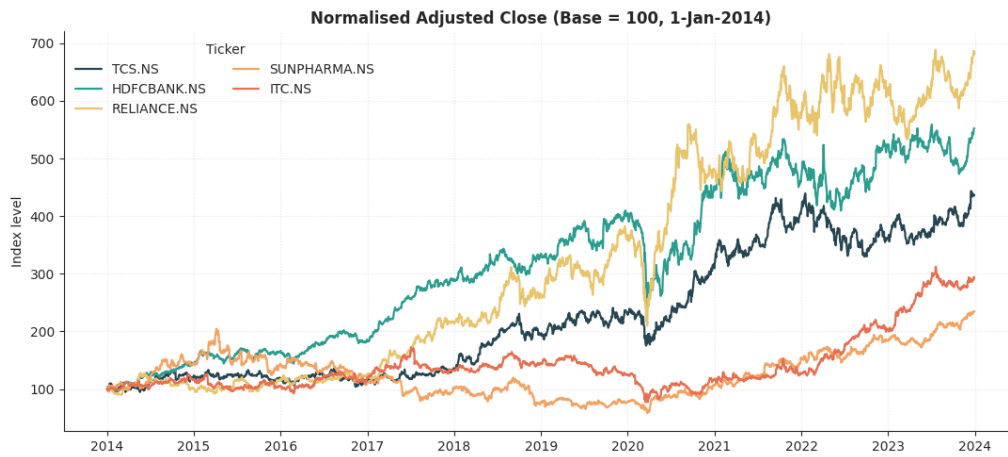


Figure 2: Normalised adjusted close prices, base = 100 on 1-Jan-2014.

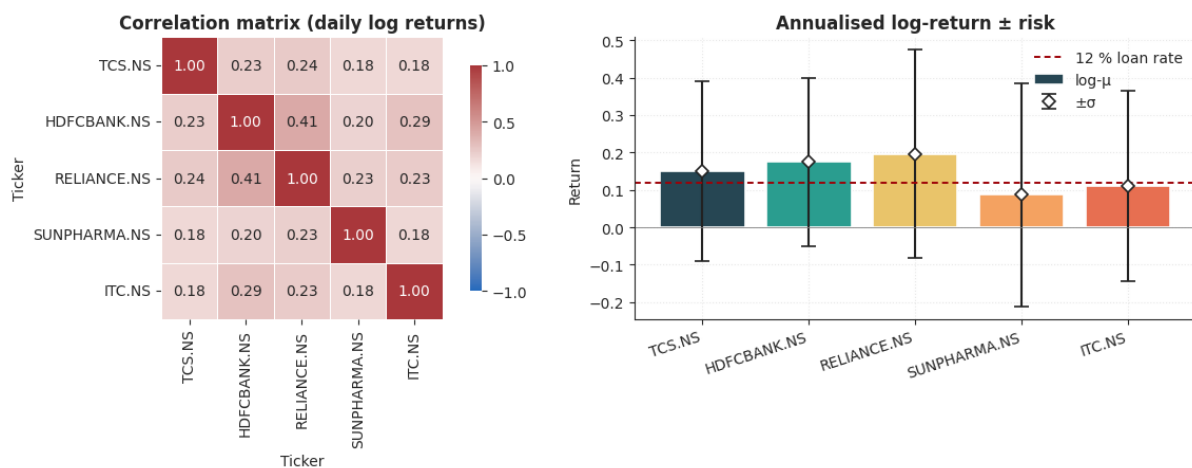


Figure 3: Left: correlation of daily log-returns. Right: annualised log-return (μ) with $\pm\sigma$ error bars; the dashed red line is the 12% effective borrow rate.

Figure 3 gives the correlation matrix and the per-stock risk–return summary. All pairwise correlations sit between 0.18 and 0.41, confirming the “uncorrelated” assumption is roughly

defensible. Three names have an annualised log-mean above the 12 % loan rate (TCS 15.0 %, HDFCBANK 17.5 %, RELIANCE 19.6 %); SUNPHARMA and ITC sit clearly below it and so are unattractive when borrowing is on the table.

4 Results

4.1 Optimal discrete portfolios

Each of the 70 portfolios is run through the Monte-Carlo engine with both goal sequences, using common random numbers (seed 42) so the comparison is free of path noise. The two winners are:

- **Sequence A:** 100 % RELIANCE, $P_{\text{success}} = 3.08\%$.
- **Sequence B:** 50 % HDFCBANK + 50 % RELIANCE, $P_{\text{success}} = 82.22\%$.

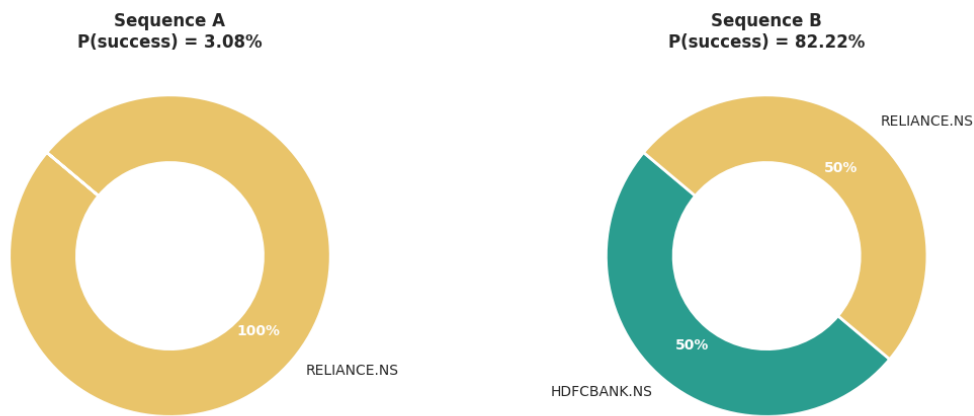


Figure 4: Optimal static allocations under each goal sequence.

Figure 5 shows the full success-probability landscape across the 70 candidate portfolios. Sequence A is brutal — the best portfolio is the only one with non-trivial success probability, and the rest cluster near zero. Sequence B is much friendlier; the top quartile of portfolios all clear 70 %.

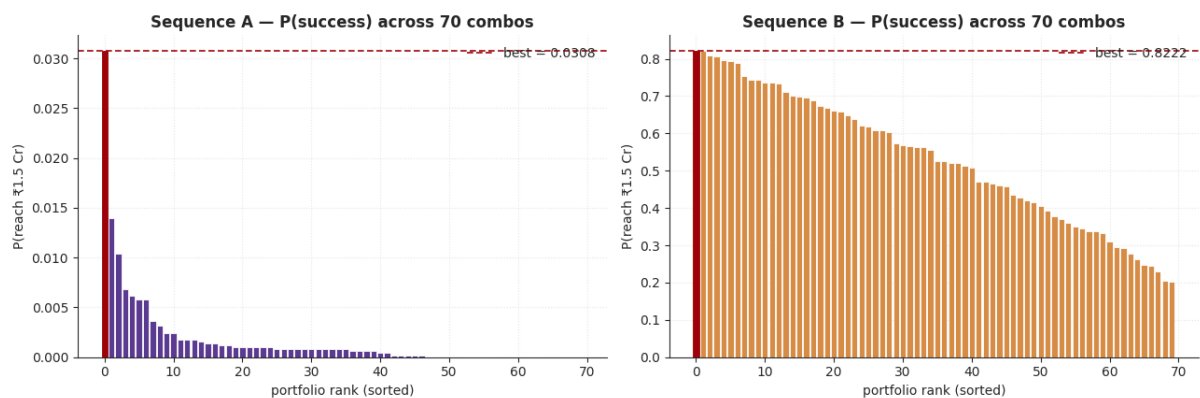


Figure 5: Sorted success probabilities across the 70 discrete portfolios. The crimson bar marks the optimum in each panel.

4.2 Behaviour of the optima

Figure 6 shows the distribution of terminal net worth across 5,000 paths under each optimum, and Figure 7 plots 200 sample trajectories of net worth over the 20-year horizon. Goal-payment kinks are visible in the path plots.

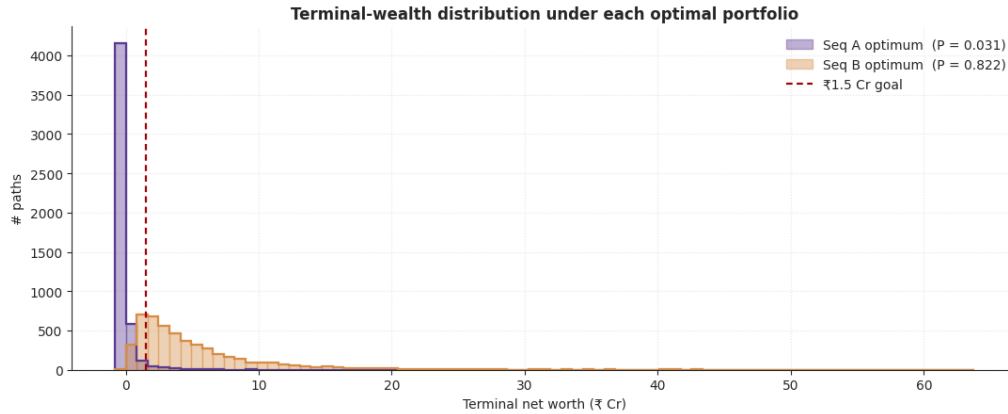


Figure 6: Terminal net-worth distribution under each optimal portfolio.

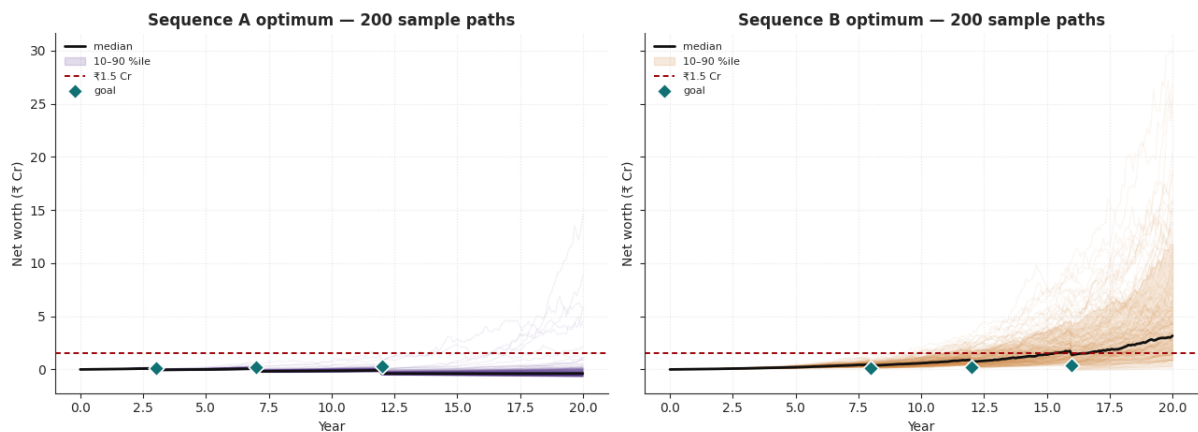


Figure 7: 200 simulated wealth trajectories under each optimal portfolio. Shaded band: 10–90 %ile; black line: median; diamonds: intermediate goals; dashed red line: 1.5 Cr terminal goal.

5 Bonus: continuous weights with short-selling

We use `scipy.optimize.basinhopping` — a global solver — wrapping `minimize` with `method="SLSQP"` as its local polish. At each iteration `basinhopping` perturbs the weights by a random kick (`stepsize = 0.4`); the inner SLSQP solve re-enforces $\sum_i w_i = 1$ exactly. The combination explores the constrained surface globally while keeping every candidate feasible:

- **Sequence A continuous:** Short ITC heavily (-1.04), Mildly short TCS and SUN-PHARMA, Lever RELIANCE ($+1.57$) with HDFCBANK ($+0.53$). $P_{\text{success}} = 28.92\%$.
- **Sequence B continuous:** Long HDFCBANK ($+0.68$) and RELIANCE ($+0.70$), mild short on ITC (-0.42). $P_{\text{success}} = 89.42\%$.

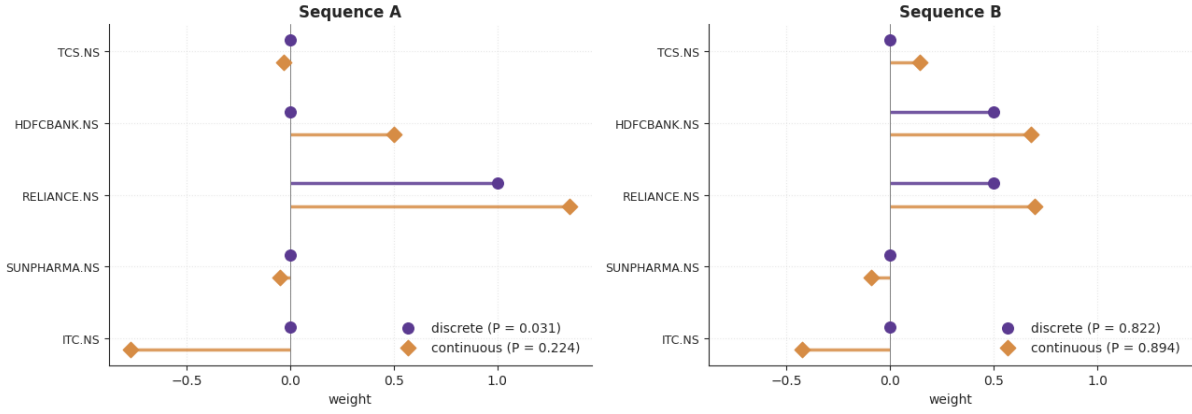


Figure 8: Discrete versus continuous optimal weights. Circles: discrete optimum; diamonds: continuous (shorts allowed).

6 Discussion

Sequence A's three early goals (Y3, Y7, Y12) land before contributions catch up, so most paths borrow at 12 %. The only way to claw back to ₹1.5 Cr is to ride the highest- μ asset, so the optimum collapses to 100 % RELIANCE. P_{success} is still only $\approx 3\%$: the budget is structurally tight approx ₹71.5 L of contributions must fund approx ₹70 L of intermediate goals *and* a ₹1.5 Cr terminal target.

Sequence B postpones every goal past year 8, by which point contributions already exceed the first goal. The borrow penalty barely fires and the problem becomes a clean mean-variance trade-off, won by the 50/50 HDFCBANK + RELIANCE blend; SUNPHARMA and ITC are excluded because their lower means aren't redeemed by the (modest) diversification benefit.

The bonus only sharpens this, With shorts and leverage allowed, the optimiser short-sells some of the lower-return names and levers RELIANCE, lifting P_{success} from 3 % to 28 % on Sequence A and from 82 % to 90 % on Sequence B. The wider lesson: the *timing* of liabilities reshapes the optimum more than their size — early goals force concentration on the highest- μ name, late goals reward diversification.

7 Conclusion

Sequence	Discrete optimum	P_{success}	Continuous P
A (early-loaded)	100 % RELIANCE	3.08 %	28.92 %
B (back-loaded)	50 % HDFC + 50 % RELIANCE	82.22 %	89.42 %

Table 1: Optimal static portfolios for each goal sequence.

Same five stocks, same 20-year horizon, but the two goal schedules push the static optimum to opposite extremes. Sequence A's early borrow penalty leaves no room for diversification, so the brute force collapses to 100 % RELIANCE (highest- μ) and clears ₹1.5 Cr only 3 % of the time. Sequence B's late goals let contributions compound first, the borrow constraint barely binds, and a 50/50 HDFC BANK + RELIANCE blend reaches 82 %. The continuous bonus (shorts on lower-return names, leverage on RELIANCE) lifts these to 28 % and 90 %, confirming that the timing of liabilities, not their size, reshapes the optimisation.